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Knowledge structure enhanced graph representation learning model for attentive knowledge tracing

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Abstract

Knowledge tracing (KT) is a fundamental personalized-tutoring technique for learners in online learning systems. Recent KT methods employ flexible deep neural network-based models that excel at this task. However, the adequacy of KT is still challenged by the sparseness of the learners' exercise data. To alleviate the sparseness problem, most of the exiting KT studies are performed at the skill-level rather than the question-level, as questions are often numerous and associated with much fewer skills. However, at the skill level, KT neglects the distinctive information related to the questions themselves and their relations. In this case, the models can imprecisely infer the learners' knowledge states and might fail to capture the long-term dependencies in the exercising sequences. In the knowledge domain, skills are naturally linked as a graph (with the edges being the prerequisite relations between pedagogical concepts). We refer to such a graph as a knowledge structure (KS). Incorporating a KS into the KT procedure can potentially resolve both the sparseness and information loss, but this avenue has been underexplored because obtaining the complete KS of a domain is challenging and labor-intensive. In this paper, we propose a novel KS-enhanced graph representation learning model for KT with an attention mechanism (KSGKT). We first explore eight methods that automatically infer the domain KS from

learner response data and integrate it into the KT procedure. Leveraging a graph representation learning model, we then obtain the question and skill embeddings from the KS-enhanced graph. To incorporate more distinctive information on the questions, we extract the cognitive question difficulty from the learning history of each learner. We then propose a convolutional representation method that fuses these distinctive features, thus obtaining a comprehensive representation of each question. These representations are input to the proposed KT model, and the long-term dependencies are handled by the attention mechanism. The model finally predicts the learner's performance on new problems. Extensive experiments conducted from six perspectives on three real-world data sets demonstrated the superiority and interpretability of our model for learner-performance modeling. Based on the KT results, we also suggest three potential applications of our model.

KEYWORDS

cognitive question difficulty, graph representation learning, intelligent tutoring systems, knowledge structure discovery, knowledge tracing, learner proficiency estimation

1 | INTRODUCTION

The popularity of online learning has increased in recent years, with increasingly many intelligent tutoring systems (ITSs) and massive open online course (MOOC) platforms becoming available to learners.^{1–3} In general, these platforms enable knowledge-acquisition through a series of individualized learning activities that accommodate learners with different needs and knowledge proficiencies. A key technique underlying these adaptive tutoring services is learner modeling.^{4,5} In this paper, we model the knowledge proficiencies of learners. This task, known as knowledge tracing (KT),^{6–12} aims to model learner performance and hence discover the latent knowledge proficiencies of learners in mastering knowledge concepts.

Knowledge tracing (KT) in an online learning system assesses the learning outcomes at the knowledge-concepts granularity in a longitudinal manner. A KT task is generally formulated as the problem of predicting learners' performance on new problems based on their learning histories^{4,6} (note that learners' current knowledge states are latent but their performances on questions based on their knowledge proficiencies are explicit and easily obtainable). As in traditional educational environments, KT evaluates a learner's knowledge through questions or quizzes, but unlike traditional learning, is based on the accumulated exercise records of learners over a long time. Based on their inferred knowledge states, learners are provided with various adaptive services that suit their

individual needs, thereby improving their learning efficiency.⁴ For this reason, KT has been a popular interdisciplinary research topic across education, computer science, and cognitive science.^{4,6,13}

Massive efforts have been devoted to skill-level KT,^{12,14} which performs KT based on the skills (or “knowledge concepts”) required in a specific domain. Each question in a KT task is correlated with one or more skills needed to solve the question (e.g., “3+5” corresponds to the skill “addition of integers”), and each skill is related to many questions. The question–skill mapping information is typically encoded as a Q-matrix of prior knowledge provided by education experts,^{11,13} and can be naturally represented as a question–skill relation graph (for an example, see Figure 1(left)). Most of the existing KT methods train the KT models on skills rather than questions, as the number of problems is far greater than the number of skills and most students only attempt a small part of the questions. Accordingly, the response data are quite sparse.^{12,14} Skill-level KT is feasible to some extent because skill mastery can largely affect the correctness of question answering. Therefore various KT methods proposed under this setting have achieved good performance.

In skill-level KT models, all questions pertaining to a specific skill are considered as the same inputs (and multiple skills corresponding to a question are merged into one new skill). This approach loses the distinctive information related to individual questions, leading to imprecise inferences of the learners’ knowledge states.^{12,14–16} For example, in Figure 1(left), the questions “3+5” and “345+6789” both require the skill “addition of integers,” and are considered as the same inputs when building the KT models, which ignores their different difficulty levels. Existing research have proved that question difficulty undoubtedly influences the learner performance,^{11,17,18} and the relative difficulty level of a specific question varies from learner to learner. The difficulty level of the same question for a specific learner also depends on the learner’s cognition, which varies over time. The existing work either neglects the question difficulty or assumes a constant difficulty level.^{7,8} In the latter case, the question difficulty is question-oriented, which is unrealistic in dynamic learning processes.¹¹ Cognitive psychologists¹⁹ have also shown that in human learning, the difficulty level of the same question also depends on the cognitive load it presents to different learners, owing to their different knowledge structures. In this paper, we refer to such user-oriented difficulty as the cognitive question difficulty, and explicitly calculate it based on the learning history of each learner. By associating the cognitive question difficulty with the question representation, we

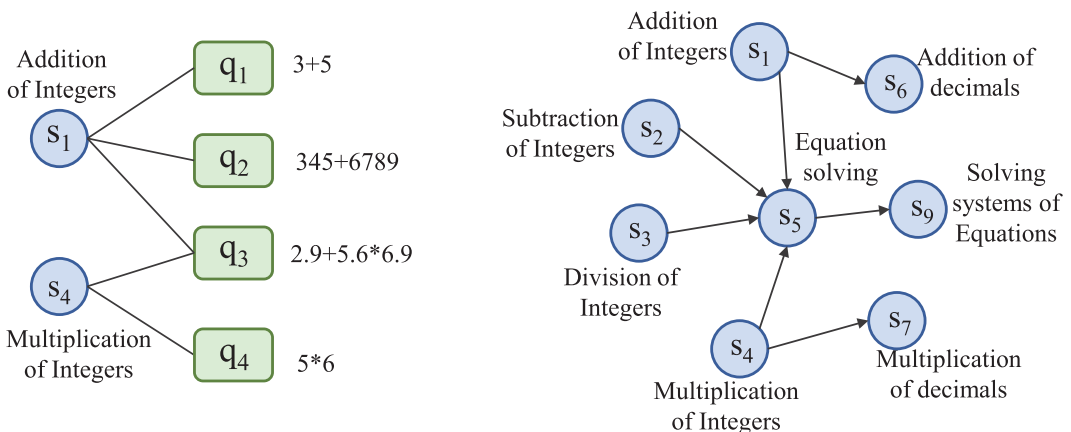


FIGURE 1 (Left) Illustration of a question–skill relation graph, and (Right) part of the knowledge structure in the algebra domain [Color figure can be viewed at wileyonlinelibrary.com]

introduce extra distinctive information for each question and improve the reliability of tracing learner proficiency over the long-term.

The interdependencies between the skills in the knowledge structure (KS) have long been acknowledged in both cognitive science and artificial intelligence.^{20,21} The prerequisites between pedagogical concepts can be represented as a knowledge graph²² (see Figure 1(right), for an example). Nevertheless, the KS (which specifies the relations among skills) has rarely been integrated in KT models because obtaining the complete KS of a domain is labor-intensive and the KS is not easily inferred from the data.^{23–25} To avoid these difficulties, most of the KT models simply assume that all questions and skills are independent. In the real world, questions are interrelated to each other and are also closely related to the underlying skills required for their solution. When learners grow their knowledge from a certain question incorporating *skill*₁, they also improve their attainment of *skill*₂ to some extent (assuming that *skill*₁ is related to *skill*₂ in the KS). For example, a learner who attempts an exercise requiring the “equation solving” skill will also deepen his or her understanding of the skill “solving systems of equations.” In this paper, we infer the KS from learners’ response data and integrate it with the KT model. Our method offers two advantages over the existing models: first, it inputs extra information into the question representation by referencing the KS (thus alleviating the data sparsity problem in the question representation), and second, it models the impact of previous experiences on future exercise during the knowledge evolution. In addition, incorporating the KS into the KT procedure can capture the long-term dependencies in the exercising sequences,²⁶ further improving the precision of inferring the dynamic knowledge proficiencies of learners.

Moreover, most of the previous work on KT represents the questions for model building using one-hot encoding.⁷ The resulting data are often too sparse to represent sufficient information for the KT task, thus leading to performance degradation.^{12,14,25,27} In recent work on graph representation learning, the model is trained on a dense embedding of the graph, which improves the performance on various tasks.^{28,29} Motivated by the high ability of graph neural networks to extract graph representation by aggregating the information from neighbors, we apply the graph representation learning method to obtain question- and skill-embedding from the KS enhanced question–skill relation graph. The learned embeddings from the graph incorporate not only the explicit question–skill relations but also the implicit question–question and skill–skill relations in the graph. We also propose a convolutional representation method that incorporates additional information and considers their interactions, thus generating dense and meaningful representations of the input questions and potentially further improving the model performance.

In summary, we propose a new KT model named Knowledge Structure enhanced Graph Knowledge Tracing (KSGKT) that traces the evolution of learners’ knowledge proficiencies based on their exercising logs in online learning systems. We first explore eight methods to infer the domain KS from the learner response data and integrate it with the original question–skill relation graph to obtain the KS enhanced question–skill relation graph. Leveraging a graph representation learning model (Metapath2Vec²⁸), we then obtain the dense question and skill embeddings from the enhanced graph. To overcome the limitations of skill-level KT models, which neglect the distinctive information related to the questions, we propose the cognitive question difficulty, which depends on the learning histories of learners, and calculate the skill- and question-level cognitive difficulties of each question. We finally propose a convolutional representation method that integrates the question and its associated skill embedding information into the multilevel cognitive question-difficulty information, thus obtaining a comprehensive representation of each attempted question. These representations for the learners’ exercising sequences are fed into the proposed KT model, which considers the long-

term dependencies using an attention mechanism, and finally predicts the learners' performance on new problems. The main contributions are listed below.

- We explore eight methods that automatically discover the domain KS from learner response data, and incorporate the result into the KT procedure. To our knowledge, we solve KT tasks using KS enhanced graphs for the first time.
- We propose a KS-enhanced graph representation learning model that learns the dense question and skill embeddings in the KS-enhanced graph.
- We propose a method that obtains the multilevel cognitive difficulty of each question, and a convolutional representation method that fuses these distinctive heterogeneous features into a comprehensive question representation. Moreover, KT is supplemented with three attention methods.
- We conduct comprehensive experimental evaluations from six perspectives on three real-world data sets. The results demonstrate the superiority and interpretability of our method in dynamically modeling the learning performance and discovering the KS from data.

Note that we solve the KT task in this paper by not only designing and enhancing the pipeline using the deep-learning techniques, but also integrating the learner's knowledge construction process and various educational theories into the deep learning framework.

Organization: The remainder of this paper is organized as follows. Section 2 presents the background information (related models and techniques), and Section 3 formulates the KT problem and introduces the definitions of this paper. The proposed KSGKT model is detailed in Section 4. Section 5 explains the experimental settings, and Section 6 presents the experimental results and analysis. Based on the KT results, Section 7 introduces three potential applications of our method. The paper concludes with Section 8.

2 | RELATED WORK

2.1 | Knowledge tracing

2.1.1 | Probabilistic and factor analysis models

Various models can dynamically obtain the long-term performance of learners and trace the learners' knowledge proficiency over time.⁴ This task was pioneered by Bayesian knowledge tracing (BKT),⁶ a probabilistic model based on the hidden Markov model (HMM) that separately tracks the proficiency of each KC without considering the contextual trial sequence of all skills and inter-skill similarity. To trace a learner's proficiency using the whole contextual sequence, researchers have proposed various factor-analysis models such as item response theory (IRT),³⁰ the additive factor model (AFM),³¹ and performance factor analysis (PFA).³² These models predefine a delicate model framework and assign the considered factors as model parameters, which are learned from the data to generalize the observations. The IRT model predicts a learner's performances from the learner's ability and the question difficulty. The AFM models the probability of successful task completion from the difficulty of the skills involved in answering the questions and the number of attempts on questions requiring those skills.³¹ PFA improves the AFM by separately considering the successful and failed attempts.³² A famous study employed a knowledge tracing machine (KTM) based on the factorization

machine, providing a generic framework for incorporating learning-related side information into a student model.⁹ The KTM model encompasses the IRT, LFA, and PFA models. Further research was conducted recently based on the KTM, for example, the item Difficulty, student Ability, Skill, and Student Skill practice History (DAS3H)^{33,34} and recurrent KTM (RKTM)³⁵ by enriching the KTM with additional information. Other researchers have incorporated side information, such as item difficulty³⁶ and uncertainty of model estimation,³⁷ into probabilistic models and factor analysis models.

2.1.2 | Deep learning models

Various neural networks have been recently used for the KT task and show significant improvements in model performance over traditional models due to their excellent abilities in conducting sequential learning on big data.^{7,8,38} The pioneering work on deep KT (DKT)⁷ obtained the learners' latent knowledge proficiency, which was extracted from the exercising sequences by a recurrent neural network (RNN) or long short-term memory (LSTM). Based on the single-state RNN, Long et al. proposed a new model named individual estimation KT (IEKT) by incorporating the learner characteristics, such as the learners' cognition levels and knowledge acquisition sensitivity on questions.³⁹ Dynamic key-value memory networks for KT (DKVMN)⁸ employed a memory-augmented neural network (MANN), which used an auxiliary memory to record the proficiency of a learner on each latent concept. Sequential key-value memory networks (SKVMNs)²⁶ modeled student learning by unifying the strengths of RNN (recurrent modeling capacity) and MANN (high memory capacity). A newly proposed network named Transformer⁴⁰ has also been adapted to student learning in various ways.³⁸ To focus on the relevant interactions, the Transformer framework applies a self-attention mechanism to the input data, and hence incorporates the inner relations in the exercise sequences into the network. Attention based KT models inspired by this study have become an active research area,^{15,16,41} a representative work is the attentive KT (AKT) model,¹⁶ which obtained the state-of-the-art performance on this task. However, in most of these models, the KT is based on the skills in a specific domain. Such skill-level KT models do not distinguish questions containing the same skills. The loss of distinctive information related to the questions may lead to imprecise inferences of the learners' knowledge states. Moreover, almost all of these models simply assume that all questions and skills are independent, which is unrealistic in the actual learning process.

2.1.3 | Graph-based models

The relations between questions and skills have been considered in various graph-based KT models.^{12,14,25,27} Liu et al.¹⁴ built a question-skill bipartite graph based on the Q-matrix in the domain and pretrained the question and skill embedding in the graph using three constraints of explicit question-skill relations and implicit relations of skill similarity and question similarity. As the dense embeddings of questions and skills contain the relations in the graph, the pretrained embedding-fed KT model outperforms the baseline models. Yang et al.¹² proposed the graph-based interaction KT (GIKT) model and incorporated the question-skill correlations via embedding propagation on the question-skill relation graph using a graph convolutional network (GCN). Sun et al.⁴² proposed a question-question collaborative embedding method for KT (CoKT) and took the learned question embeddings based on the question similarity as input.

Tong et al.²⁷ introduced problem schema with a hierarchical exercise graph to KT. The problem schema clustered similar exercises into the same group to incorporate the question–question relations. Pandey et al.¹⁵ calculated the exercise–exercise relations based on the performance data and the exercise text, and incorporated it into a transformer model for relation-aware KT. All of these methods exploit the relation information in various graphs and introduce the distinctive information of questions into the KT task. However, they ignore the KS information of the skills in the domain. Similarly to our work, Nakagawa et al.²⁵ and Chen et al.²⁰ incorporated the KS information into the KT procedure. Chen et al.'s work assumes that the KS is already given by experts, and models the prerequisite as an ordering pair. In this way, the mastery of related skills is constrained by referring to the KS. Nakagawa et al. used the KS information and a graph neural network (GNN) to update the learners' hidden knowledge states. Unlike these earlier works, our method learns the KS from learner response data and integrates it with the Q-matrix to build a KS-enhanced relation graph. Leveraging the Metapath2Vec method, we learn the dense embeddings of all nodes, which incorporate the multihop question–question, question–skill, and skill–skill relations in the domain. The different types of models are compared in Table 1.

2.2 | Knowledge structure discovery

Knowledge structure discovery facilitates the intelligence of online learning systems by providing personalized learning paths and services to accommodate the needs of different learners. Automating this process has been attempted in several studies. Most of the existing KS-discovery methods identify the latent skills required for answering the questions, and find the similarities among the questions in the domain for clustering the potential KS. The technique of Lindsey et al.⁴³ discovers a skill set that learners use for problem solving. This technique assumes a monotonic improvement of learner performance along a sequence of questions requiring the same skill. Pan et al.²³ automatically inferred the prerequisite relation between knowledge concepts in MOOCs using natural language processing techniques. Mussack et al.⁴⁴ discovered the problem similarity by combining the problem features and learner behaviors. Rihák et al.⁴⁵ and Nazaretsky et al.⁴⁶ proposed various measurement indices that cluster the education items from performance data. Their methods assume that learner performances are similar on items requiring the same skills. However, in these studies, clusters of questions requiring the same latent skills are regarded as the KS in the domain; in contrast, our KS is constructed from the prerequisite relations among explicit skills.

Wang et al.⁴⁷ proposed a latent-variable selection method with regularization for cognitive diagnostics, which learns the skill hierarchies from learner response data. Using a DKT model, Zhang et al.²⁴ discovered the topological order of skills from learners' exercise performance. Like our group, they intuited that the KS is difficult to directly extract from response data, but this difficulty can be bridged by ordering the learners' mastery of skills. Hence, in this paper, we discover the prerequisite skill pairs in the KS by ordering the learners' performances in question-solving (i.e., their mastery of the underlying skills).

2.3 | Graph representation learning models

Graph representation learning models such as GNNs have solved various tasks through their excellent ability to process graph-structure data. Traditional machine learning methods have

TABLE 1 Comparisons of different learning models

Category	Data source						Consider sequence order
	Representative work	Typical technique	Problem/skill representation	Difficulty	Q-S relation	Elapsed KS time	
Probabilistic model	BKT, FuzzyCDM	HMM, Bayesian Network	Parameters	-	×	×	✓ Single
Factor analysis model	IRT, AFM, PFA, KTM, DAS3H, RKTM	Linear/logistic regression	Parameters	Skill-level constant	×	×	×
Deep learning model	DKT, DKVMN, SAINT, AKT, IEKT	RNN, LSTM, MANN, Transformer	One hot encoding	-	×	×	✓ Single
Graph-based model	GIKT, PEBG, RKT, HGKT, CoKT	GCN, GNN	Dense embedding	Constant	✓	×	Multiple ✓
Our model		Metapath2Vec	Convolutional dense embedding	Multilevel, adaptive	✓	✓	Multiple ✓

limited ability to extract and encode the high-dimensional, non-Euclidean information on graph structure from graphs.⁴⁸ Following the edges in the graph, GNNs obtain the node representation from a whole graph by propagating and aggregating information from the neighbor nodes. In this way, node embeddings can summarize their graph positions and the structures of their local-graph neighborhoods.⁴⁸ The obtained graph-based embeddings can be directly used for various downstream tasks. In the deep-learning models of KT tasks, they alleviate the sparsity problem of the one-hot representation. One widely used GNN model is the GCN, which was recently used as the input-question representation of KT.^{12,27} GCNs are especially suitable for isomorphic graphs, in which the nodes are of the same type. Recent graph-to-vector models such as Metapath2Vec²⁸ have proven successful in heterogeneous graph-representation learning. Such models are eminently suitable for KT tasks because question-skill graphs are typical heterogeneous graphs. In this paper we leverage the Metapath2Vec model to learn the dense embedding from the relation graph, and input the obtained embedding to the model that traces the evolution of learner proficiency.

3 | PROBLEM FORMULATION

Let us suppose that in a learning system, each question is associated with one or several skills needed to solve the question. This association is generally given as a binary q -matrix O in which each entry $O_{j,k} \in \{0, 1\}$ indicates whether the j th question is related to the k th skill. Learners can learn each skill by attempting several corresponding questions. Moreover, each learner can individually answer the questions at different times, and will answer a question correctly or incorrectly at each attempt. Whenever a learner attempts an item at time point t , the system generates an exercise record $n_t = (u_i, q_t, s_k, et_t, r_t)$, where n_t is a quintuple including the learner ID $u_i \in \{u_1, u_2, \dots, u_I\}$, exercise ID $q_t \in \{q_1, q_2, \dots, q_J\}$, underlying skills $s_k \in \{s_1, s_2, \dots, s_K\}$, the elapsed time et_t spent on solving the given question q_t , and the correctness $r_t \in \{0, 1\}$ of the learner's answer at timestamp t . A learner's exercising process is then modeled as a sequence $X = \{n_1, n_2, \dots, n_T\}$, where T indicates the latest timestamp.

Given a learner's past exercising sequence X in a system and a new question q_{t+1} , our goal is to model the learner-question interaction procedure, and hence track the evolution of a learner's knowledge proficiency over time. Based on the learner knowledge proficiency, we then predict the probability $p(r_{t+1} = 1 | X, q_{t+1})$ that the learner answers question q_{t+1} correctly.

We now introduce some definitions that will be used in the following sections.

Definition 1 (Question-skill graph). The q -matrix O containing the question-skill relations can be naturally represented as a question-skill graph $G = \{Q, S, L\}$, where Q and S are sets of question and skill nodes, respectively, and $L = [O_{jk}] \in \{0, 1\}$ indicates whether question node q_j and skill node s_k are connected by an edge.

Definition 2 (Knowledge structure). The KS is a directed graph with all skills as nodes. It can be represented as $KS = \{S, \vec{L}\}$, where $\vec{L}$ denotes the prerequisite relations between two skill nodes in the graph.

Definition 3 (Cognitive question difficulty). This factor models the learner-oriented question difficulty in the dynamic learning process based on the learners' current

cognition (i.e., their current cognitive structures built during the exercising solving procedure).

In this paper, the incorrect interactions in the learners' previous exercising process indicate the cognitive difficulty of answering the questions. Moreover, learners attempt the same question using their skills at different frequencies. To model this behavior, we define the cognitive question difficulty of each question at both the question and skill levels, as follows:

$$\Psi_{v,t|v=\{q,s\}} = \begin{cases} \left\lceil \frac{|\{l_v == 0\}|_{0:t}}{|N_v|_{0:t}} * (c - 1) \right\rceil, & \text{if } |N_v|_{0:t} \geq 5 \\ c, & \text{otherwise} \end{cases} \quad (1)$$

where $\Psi_{q,t}$ and $\Psi_{s,t}$ denote the learner's cognitive difficulty at the question and skill levels at timestamp t , respectively. These entities are quantified into $c + 1$ levels, ranging from zero to c . N_v is the set of question or skills attempted by the learner before time t and l_v is the outcome of the previous attempt on the same question or skill. A zero outcome indicates a failure. If a student has attempted a question or a skill fewer than five times, the level is set to c , indicating the highest level of cognitive difficulty.

A list of symbol notations used in this study is presented in Table 2 to facilitate reading.

Algorithm 1 The proposed KSGKT method for attentive knowledge tracing

Input: the learners' exercise logs E in a system, the Q-matrix O

Output: the predicted probability p_{t+1} that the learner answers exercise q_{t+1} correctly, the matrix representation R^w of the KS, and the final embedding matrix Q, S of the question and skill nodes.

- 1: $R^w, \hat{R} \leftarrow \mathbf{0}, \alpha_{i,t+1}, p_{t+1} \leftarrow 0$;
 - 2: initialize Q, S, D from a Gaussian distribution;
 - 3: infer R^w from the exercise logs E ; ▷ Eight methods in Section 4.1
 - 4: calculate $\hat{R}$ using Equation (4) and (5);
 - 5: calculate $\Psi_{q,t}$ and $\Psi_{s,t}$ using Equation (1); ▷ Cognitive item difficulty
 - 6: **repeat:**
 - 7: $Q, S \leftarrow \text{Metapath2Vec}(\hat{R}, \text{"QSQ"}, Q, S)$; ▷ Embedding learning in Section 4.2
 - 8: $\tilde{q} \leftarrow \text{Convolution}(q = Q_q, s = \text{avg}(S_q), d_q = \Psi_{q,t}D, d_s = \Psi_{s,t}D)$; ▷ Convolutional question representation in 4.3
 - 9: $h_t \leftarrow \text{LSTM}(\text{Concat}(\tilde{q}, t_t, r_t), h_{t-1}; \theta)$; ▷ update learner knowledge state using LSTM in Section 4.4
 - 10: calculate attention $\alpha_{i,t+1}$ using Equation (15);
 - 11: $p_{t+1} = \text{MLP}(\overline{q}_{t+1}, h_t, \alpha_{i,t+1})$; ▷ Predict learner performance with three attention methods in Section 4.4
 - 12: **until** $\mathcal{L} = -\sum_t (r_{t+1} \log p_{t+1} + (1 - r_{t+1}) \log (1 - p_{t+1}))$ is minimum ▷ minimize cross-entropy loss
 - 13: return p_{t+1}, R^w, Q, S .
-

TABLE 2 A list of symbol notations used in this study

Var.	Description
u_i	Learner u_i
q_t	Question q_t
s_k	Skills s_k
r_t	The correctness of the learner's answer at timestamp t
et_t	The elapsed time spent on solving the given question at timestamp t
O	The binary q-matrix
R^w	The representation of the KS as a skill relation matrix
$\hat{R}$	Matrix representation of the KS-enhanced question-skill graph
q	Question embedding vector
$\bar{s}$	Average skill embedding vector in a question
d_q	Question difficulty embedding vector at the question level
d_s	Question difficulty embedding vector at the skill level
$\tilde{q}$	The final comprehensive question embedding
x_t	The interaction embedding at timestamp t
h_t	The hidden knowledge state of a learner at step t
$\alpha_{i,t+1}$	The attention between current questions and previous question
$g(\tilde{q}_i, \tilde{q}_{t+1})$	The correlation between current questions and previous question
p_{t+1}	The predicted learner performance at step $t + 1$

4 | PROPOSED METHOD

This section introduces the proposed KSGKT framework for KT, as shown in Figure 2. The framework proceeds the KT task using five modules: knowledge structure discovery, graph-based embedding learning, convolutional question representation, attention mechanism, and learner knowledge state evolution. Before conducting the embedding learning, we must build the KS-enhanced question-skill graph by integrating the KS inferred from the learner response data into the question-skill graph. Based on the built KS-enhanced question-skill graph, the dense embeddings of all skill and question nodes are obtained through the graph embedding learning method Methpath2Vec. To incorporate more distinctive information of the questions, the cognitive question difficulty at both the question and skill levels are also inferred from the learning histories of individual learners. A convolutional representation method is then proposed to fuse the question and skill embeddings with the cognitive question difficulty. It considers each separate factor and the interactions between each pair of factors, thus obtaining the comprehensive representations of questions (Figure 3). These representations are fed to the attentive KT network for predicting learner performance (see Figure 4, for the KT procedure). To capture the long-term dependencies in the exercising sequence, different attention mechanisms are applied. For reference, the algorithm for our proposed method is shown in Algorithm 1.

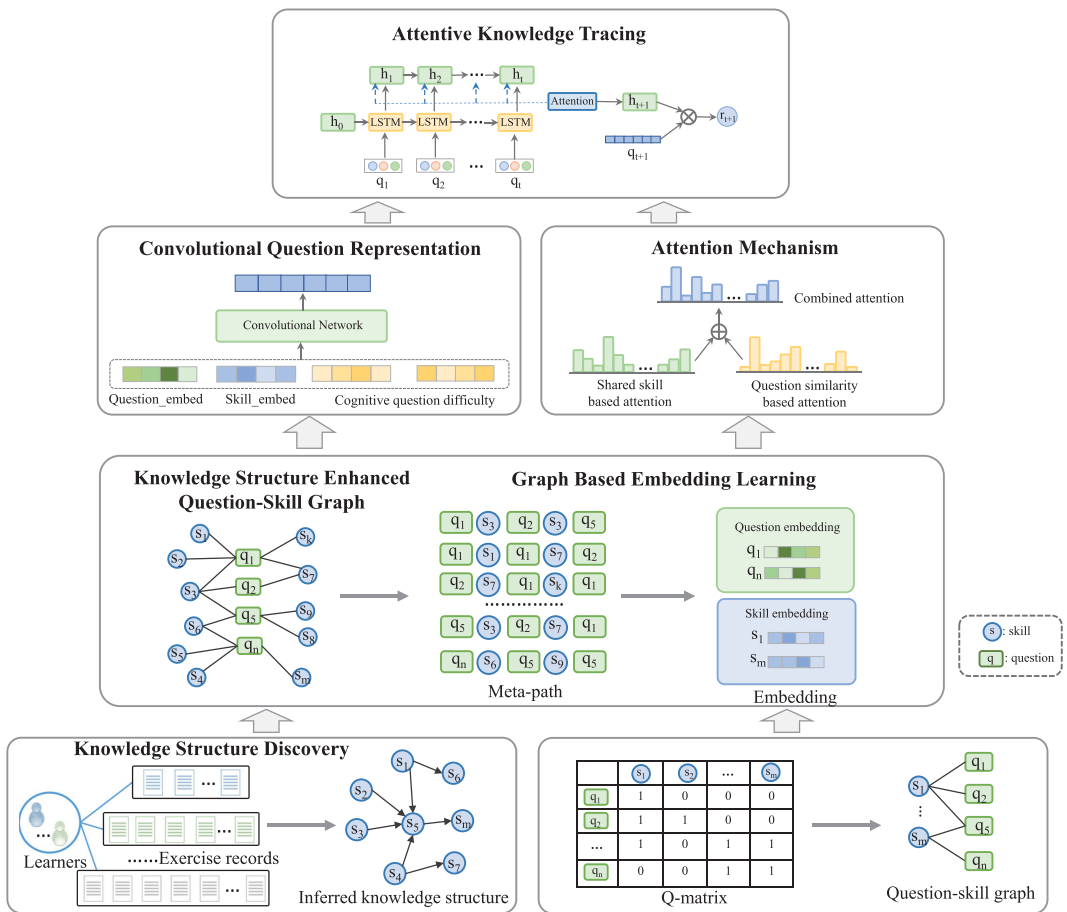


FIGURE 2 The proposed KSGKT framework for attentive knowledge tracing [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.27763)]

4.1 | Inferring KS from data

In practical educational scenarios, there always exists a topological order (KS) among the skills in a domain, because skills are taught and learned in sequence. In many learning systems the KS information is never provided and must be obtained by time-consuming labor. We observed that if learners have not mastered a specific skill, their probability of incorrectly answering questions requiring the prerequisite skills will increase. Moreover, learners' mastery of a skill can be indicated by their performances on the attempted questions requiring that skill. Based on these observations, this paper aims to discover the KS from the learner response data.

Here we infer the KS from the order of the learners' mastery of skills, which is explicitly represented by the exercising performance data. We first build a skill relation matrix $R \in \mathbb{R}^{|S| \times |S|}$, in which entry $R_{i,j}$ represents the prerequisite relation $s_i \rightarrow s_j$ between skill s_i and s_j . Inspired by the definition of question similarity in previous methods,^{27,46,49} we explore the underlying KS using eight methods: Skill Transition, Cohen's Kappa, Adjusted Kappa, Phi

coefficient, Yule coefficient, Ochiai coefficient, Sokal coefficient, Jaccard coefficient, as described below.

- *Skill transition*: The skill-transition matrix R contains the transitions of different skills. Its entries are $R_{ij}^{SK} = \frac{n_{ij}}{\sum_{k=1}^{|K|} n_{i,k}}$, where n_{ij} denotes the number of times in which skill s_j is trained immediately after training skill s_i .

To further leverage the impact of the learners' performance of one skill on the performance of another, we summarized the learners' performance on skill pair s_i and s_j in a contingency table (Table 3). As mentioned above, we interpreted the learners' correct or incorrect responses as mastery indicators of the underlying skills of the given questions. It is worth noting that in Table 3, the question requiring skill s_i occurs before the question requiring s_j in the learning sequence. When there are multiple occurrences of question pairs in the learning sequence, we consider only the latest occurrence. Based on the contingency table, we discovered the KS using the evaluation indices⁴⁹ in Table 4, which measure the agreement of the prerequisite relation between a pair of skills. These indices are widely used for measuring associations between two variables.

As the KS is always a unidirectional graph, we simplified it by a suitable strategy. We also imposed a threshold that controlled the sparsity of the relations in KS. The final skill relation matrix was denoted as R^w , $w \in \{SK, Kappa, Kappa', Phi, Yule, Ochiai, Sokal, Jaccard\}$. The elements along the diagonal of R^w were set to one.

$$\begin{cases} R_{ij}^w = \max(R_{ij}^w, R_{ji}^w), R_{ji}^w = 0, & \text{if } R_{ij}^w \geq R_{ji}^w \\ R_{ji}^w = \max(R_{ij}^w, R_{ji}^w), R_{ij}^w = 0, & \text{otherwise} \end{cases} \quad (2)$$

TABLE 3 Contingency table for a pair of skills s_i and s_j

	s_j master	s_j not master	Total
s_i master	a	b	$a+b$
s_i not master	c	d	$c+d$
Total	$a+c$	$b+d$	$a+b+c+d$

TABLE 4 Evaluation indices for obtaining KS using the contingency table

Cohen's Kappa	$R_{ij}^{Kappa} = 2(ad - bc) / \{(a + b)(b + d) + (a + c)(c + d)\}$
Adjusted Kappa	$R_{ij}^{Kappa'} = 2(ad - bc) / \{(a + c)(c + d)\}$
Phi coefficient	$R_{ij}^{Phi} = (ad - bc) / \sqrt{(a + b)(b + d)(a + c)(c + d)}$
Yule coefficient	$R_{ij}^{Yule} = (ad - bc) / (ad + bc)$
Ochiai coefficient	$R_{ij}^{Ochiai} = a / \sqrt{(a + b)(a + c)}$
Sokal coefficient	$R_{ij}^{Sokal} = (a + d) / (a + b + c + d)$
Jaccard coefficient	$R_{ij}^{Jaccard} = a / (a + b + c)$

$$R_{ij}^w = \begin{cases} 1, & \text{if } R_{ij}^w \geq \text{threshold} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

4.2 | Embedding learning on the KS-enhanced question–skill graph

After obtaining the inferred skill relation matrix $R^w \in \mathbb{R}^{|S| \times |S|}$ of the KS and the q-matrix $O \in \mathbb{R}^{|Q| \times |S|}$ for the question–skill graph, we built a KS-enhanced question–skill graph and conducted embedding learning on it.

The matrix representation $\hat{R} \in \mathbb{R}^{|Q| \times |S|}$ of the KS-enhanced question–skill graph is obtained as follows:

$$\hat{R} = O(R^w)^T \quad (4)$$

$$\hat{R}_{ij} = \begin{cases} 1, & \text{if } \hat{R}_{ij} \geq 1 \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

In Equation (4), the transpose of the skill relation matrix R^w accounts for the skills that are prerequisite to the skills of the current question. In other words, a question requiring a specific skill is also related to the prerequisite skills.

The matrix $\hat{R}$ can be naturally represented as a graph, hereafter denoted as the KS-enhanced question–skill graph. This graph includes not only the multihop relations between questions and skills, but also the prerequisite relations among the skills. The embeddings in this graph were obtained using the Metapath2Vec method.²⁸

The Metapath2Vec method proceeds in two main steps: meta-path generation and skip-gram-based embedding learning. A meta-path is a sequence of nodes following the edges in the graph. To assure that all questions and skills appear in the final embeddings, we generate the meta-paths from the KS-enhanced question–skill graph using a question–skill–question (QSQ) pattern, in which every meta-path begins with a question node followed by a skill node and then by a question node; for example, $\rho: q_1 \xrightarrow{\hat{R}_{1,1}} s_1 \xrightarrow{\hat{R}_{2,1}} q_2 \xrightarrow{\hat{R}_{2,2}} s_2 \xrightarrow{\hat{R}_{3,2}} q_3$. Setting two hyper-parameters—the path length $\wp$ and number of paths $\aleph$ —for each question node, we generated all meta-paths on the graph. The probability of moving one step between two nodes v_i and v_{i+1} along path ρ is given by

$$P(v_{i+1}|v_i, \rho) = \begin{cases} 1/|N_i(v_i)|, & \text{if } v_{i+1} \in N_i(v_i), \text{type}(v_i, v_{i+1}) \in \text{“QSQ”} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where $N_i(v_i)$ is the set of one-hop neighbors of nodes v_i following the QSQ pattern.

Following, [28] we apply the heterogeneous skip-gram and learn the node embeddings by maximizing the probability of having context $N_i(v_i)$ given a node v_i :

$$\operatorname{argmax}_{\theta} \sum_{v \in V} \sum_{\text{type} \in \text{“QSQ”}} \sum_{c_i \in N_i(v_i)} \log p(c_i|v_i, \theta) \quad (7)$$

where $p(c_i|v_i, \theta)$ usually defined as a softmax function based on the learned embedding of nodes. Finally, we obtain all embeddings with the same dimension d of question and skill nodes in the graph, which also enclose the relation information.

4.3 | Convolutional question representation

In this section, we fuse the various feature representations through convolutional operations to obtain the comprehensive question embedding with distinctive information.

We have now obtained the distinctive embedding, contained skill embedding, and difficulty information of each question. To simultaneously preserve all of these parts, we fuse them into comprehensive question embeddings. Following [14] and [50] that learn the high-order latent patterns through feature interactions and convolution operations (rather than directly concatenating the features), we map and fuse the separate features and their interactions using convolution operations, as shown in Figure 3.

The question-difficulty information is represented as vectors using an embedding matrix D (of size $(c + 1) \times d$). The continuous embedding vectors at the question and skill levels of each question are defined as $d_q = \Psi_{q,t}D$ and $d_s = \Psi_{s,t}D$, respectively. For questions containing multiple skills, we represent the skill as the average skill embedding as follows:

$$\bar{s} = \frac{1}{|s_q|} \sum_{s_i \in s_q} s_i. \quad (8)$$

Fusing the above-obtained features, we generate the linear information M and quadratic information N for question q .

$$\begin{aligned} M &= [q, \bar{s}, d_q, d_s] \in \mathbb{R}^{4 \times d}, \\ N &= [\langle M_i, M_j \rangle] \in \mathbb{R}^{4 \times 4}, \end{aligned} \quad (9)$$

where $\langle \cdot \rangle$ represents the interactions of two vectors obtained by the inner product. We then apply the two-dimensional convolution operation with eight kernels of size 2×2 on both M and N , and maxpooling on each feature map to obtain $l_m \in \mathbb{R}^{1 \times (d-1) \times 8}$ and $l_n \in \mathbb{R}^{1 \times 3 \times 8}$. These two parts are then concatenated into eight longer vectors $l_{m+n} \in \mathbb{R}^{1 \times (d+2) \times 8}$ including the convolutions from the separate features and their interactions. Inspired by the multi-head mechanism in the transformer model,⁴⁰ we concatenate and linearly transform the eight vectors and hence obtain the final question representation $\tilde{q} \in \mathbb{R}^{1 \times d'}$.

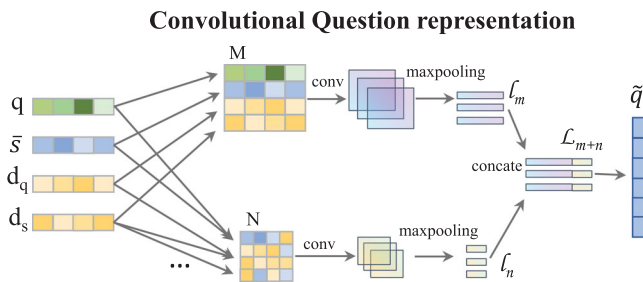


FIGURE 3 Use of convolution to fuse various distinctive features and their interactions into the comprehensive question representation [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.27763)]

$$\begin{aligned}
 l_m &= \text{MaxPooling}(\text{Conv}(M)), \\
 l_n &= \text{MaxPooling}(\text{Conv}(N)), \\
 l_{m+n} &= \text{Concat}(l_m, l_n),
 \end{aligned} \tag{10}$$

$$\tilde{q} = \text{Concat}(l_{m+n}^1, \dots, l_{m+n}^s) W^O, \tag{11}$$

where $W_O \in \mathbb{R}^{((d+2) \times 8) \times d'}$ is the parameter that transforms the convolution results into a vector.

4.4 | Learner knowledge state evolution

The learner exercising sequences are fed into an attentive KT framework that predicts the learner performance, as shown in Figure 4.

The log data of each interaction in the exercising sequences consists of a tuple representing the question, the correctness, and the elapsed time. Look-up operations are performed on an embedding matrix $E_r \in \mathbb{R}^{2 \times d'}$, in which row vector r_i contains the incorrectness or correctness of the responses. The elapsed time et strongly evidence a student's proficiency in knowledge and skills.⁵¹ This time is converted to seconds and capped at 500 s. A d' -dimensional latent embedding vector for et_k is computed as $t_k = et_k W_{et} + b_{et}$, where W_{et} and b_{et} are learnable vectors. The interaction embedding is obtained as follows:

$$x_t = \text{Concat}(\tilde{q}, t_i, r_i). \tag{12}$$

The sequence data of the learners' exercising process are modeled using LSTM⁷ within the KT framework, and the learners' knowledge states are traced at each time. Since KT problem can be regarded as a prediction problem based on the long-term time-series data, LSTM is especially suitable for such kind of task and can extract useful information in the previous learning histories to predict future performance. Actually some other kinds of RNN models,

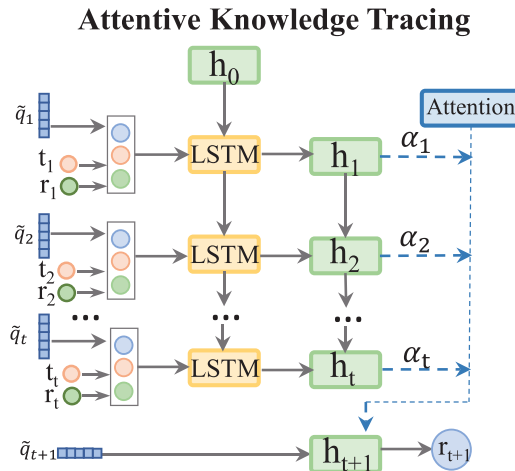


FIGURE 4 Tracking the evolution of a learner's knowledge state by attentive knowledge tracing on the exercising sequences. LSTM, long short-term memory [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.22753)]

such as the vanilla RNN, and gated recurrent unit cell (GRU) models⁵² can be also used for this task. To better compare with other KT methods and show the effectiveness of the knowledge structure enhanced graph representation learning model, here we also adopt the LSTM in the final part of our framework for fair consideration. The hidden knowledge state h_t of a learner at step t is updated based on the current input and the previous state $h_t = \text{LSTM}(x_t, h_{t-1}; \theta)$ as follows:

$$\begin{aligned} i_t &= \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i), \\ f_t &= \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f), \\ o_t &= \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o), \\ c_t &= f_t c_{t-1} + i_t \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c), \\ h_t &= o_t \tanh(c_t), \end{aligned} \quad (13)$$

where the i_t, f_t, o_t are the input, forget, and output gates, respectively, c_t is the cell memory vector, and W_* and b_* are network parameters.

We then employ an attention mechanism that accounts for the impact of previous attempts on the current attempt. A new question will likely be strongly affected by similar questions or questions requiring the same skillset as the new question. Moreover, according to forgetting-curve theory, the impact exponentially decays as time passes.⁵³ To describe these effects, we assume that the learner-knowledge state in the current step is the weighted sum of the aggregated states in the previous steps. The weights are based on the correlations:

$$h_{t+1} = \sum_{i=1}^t \alpha_{i,t+1} h_i. \quad (14)$$

The attention was calculated by three methods: shared skill-based attention, question similarity-based attention, and a combination of both former attention methods.

$$\begin{aligned} \alpha_{i,t+1} &= \frac{\exp(\text{corre}_{i,t+1})}{\sum_j \exp(\text{corre}_{j,t+1})}, \\ \text{corre}_{i,t+1} &= \exp(-\theta |t_{t+1} - t_i|) g(\tilde{q}_i, \tilde{q}_{t+1}), \end{aligned} \quad (15)$$

where $|t_{t+1} - t_i|$ is the time interval between step i and $t + 1$, $\theta > 0$ is the learnable decay rate over time, the exponential term down-weights the importance of questions in the distant past, and $g(\cdot)$ denotes the correlation between two questions. The $g(\cdot)$ s computed by the three methods provide three kinds of attentions.

- *Shared skill based-attention*: This method calculates the number of skills shared by two questions in the KS-enhanced q-matrix $\hat{R}$. We define $g(\tilde{q}_i, \tilde{q}_{t+1}) = \frac{n}{|s_{q_{t+1}}|}$, where n is the number of shared skills and $|s_{q_{t+1}}|$ denotes the number of skills contained in question q_{t+1} .
- *Question similarity-based attention*: This method calculates the correlations between two question representations by determining their similarities. We define $g(\tilde{q}_i, \tilde{q}_{t+1}) = \cos(\tilde{q}_i, \tilde{q}_{t+1})$, that is, the cosine similarities between the two vectors of question representation.

- **Combined attention:** This method combines the above correlations with coefficients. We define $g(\tilde{q}_i, \tilde{q}_{t+1}) = \lambda \frac{n}{|s_{q_{t+1}}|} + (1 - \lambda) \cos(\tilde{q}_i, \tilde{q}_{t+1})$, where λ is a tunable parameter that balances the above two correlations.

The learner performance at step $t + 1$ can be predicted from the question representation $\tilde{q}_{t+1}$ and the current knowledge state h_{t+1} as follows:

$$\begin{aligned} s_{t+1} &= \tanh(W_s[\tilde{q}_{t+1}, h_{t+1}]) + b_s, \\ p_{t+1} &= \sigma(W_p s_{t+1} + b_p), \end{aligned} \quad (16)$$

where W_* and b_* are parameters in the fully connected layer and the sigmoid activation layer, respectively.

Finally, we optimized our model by the cross-entropy loss; specifically, we minimized the following objective function between the true answer l_t and the predicted performance p_{t+1} at each interaction:

$$\mathcal{L} = -\sum_t (r_{t+1} \log p_{t+1} + (1 - r_{t+1}) \log(1 - p_{t+1})). \quad (17)$$

5 | EXPERIMENTAL SETTINGS

In a series of experimental tasks, we evaluated the proposed KSGKT model on three public real-world data sets. This section describes the experimental settings (aims, data sets, comparison baselines, setup and implementation, and evaluation metrics). The detailed experimental results and model analysis are presented in the next section.

5.1 | Experimental aims

Our experiments aimed to answer the following questions:

1. Based on the knowledge proficiency inferred from the learners' exercise histories, how well does KSGKT predict the learners' performance on new questions? (See Section 6.1).
2. How does embedding learning on the KS-enhanced graph affect the performance of the proposed model? (See Section 6.2.1).
3. Does the question embedding learned on the graph provides meaningful information? (See Section 6.2.2).
4. How well do the eight methods infer the KS from data? Is the inferred KS explainable? (See Section 6.2.3 and 6.2.4).
5. How effective is the convolutional question representation of the KT task? (See Section 6.2.5).
6. How effective is the attention mechanism in the proposed method? What is the effect of changing the attention-calculating method? (See Section 6.2.6).

5.2 | Data sets

Experiments were performed on three well-established data sets, namely, Assist0910,^{*} Assist1213,[†] and EdNet[‡]. All data sets are public real-world data sets containing the temporal interaction records between learners and real computer-aided tutoring systems. These data sets incorporate rich information on the learners and questions, along with their interactions over time; hence, they are very suitable for long-term dynamic KT studies in large-scale scenarios.

Assist0910: This data set was collected from 2009 to 2010 using the ASSISTment system.⁵⁴ It contains the interactions of 3002 learners and 17,705 items associated with 123 skills, forming 277,540 interaction records.

Assist1213: This data set, which updates Assist0910, was collected over the whole year of 2012. In this data set, 22,339 learners in the system attempted 52,825 mathematics questions requiring 265 skills. This data set contains the largest number of learners, questions, and interaction entries (nearly 2.7 million). It is a one-skill-per-question data set, meaning that each question requires one skill.

EdNet: The EdNet data set is newly available to the public.⁵⁵ Following [12], we randomly selected 5000 students who answered 12,372 questions requiring 188 skills, thus obtaining 347,866 interaction logs. This data set contains the minimum number of attempted items per learner (69.57) but the largest number of skills per item among the three data sets.

As done in existing studies,^{9,12} we eliminated noise by deleting users with fewer than 10 interaction entries and questions with “not-a number” (NaN) skills from the three data sets. Table 5 summarizes the statistics of the data sets.

5.3 | Compared models

As part of our model evaluation, we competed the model against several state-of-the-art skill- and question-based KT models. The performances of the various models were compared.

- *BKT*⁶ conducts KT using a hidden Markov model, and represents the learner knowledge states as a set of binary variables.
- *DKT*⁷ is the first KT model based on a deep neural network. This model treats the learner-knowledge prediction as a sequence learning task and captures the complex representations

TABLE 5 Statistics of the three data sets used in this study

Data set	Assist0910	Assist1213	Ednet
# of learners	3002	22,339	5000
# of questions	17,705	52,825	12,372
# of skills	123	265	188
# of interaction records	277,540	2,672,532	347,864
# skills per item	1.20	1.00	2.28
# questions per skill	172.33	199.34	149.78
# of attempted items per learner	92.45	119.64	69.57

of student knowledge using the hidden vectors of RNNs. The input is a one-hot encoding of skills.

- *DKVMN*⁸ establishes the learners' knowledge states using an auxiliary memory that augments the neural networks. This method embeds the skill information into a key matrix and accumulates the temporal information from the learners' exercising sequences. It then infers their knowledge states on these skills.
- *KTM*⁹ is a factor analysis model based on the factorization machine. It is a generic framework that incorporates the side information into the student model. Learner performance is predicted based on a sparse set of weights applied to all features in the samples.
- *DKT-Q* is a variant of DKT that replaces the skills embedding with a one-hot encoding of questions as the input.
- *DKT-Q&S* is a variant of DKT that inputs both the questions and skill representations to the DKT.
- *DKT-CQE* is a variant of DKT that inputs our convolutional question embeddings to the DKT.
- *GKT*¹² is a graph-based interaction model that learns the question representations from the question-skill relation graph using a GCN and conducts KT within the LSTM framework.
- *DAS3H*³³ is a factor analysis model built on the KTM, it considers item difficulty, student ability, skill, and student skill practice history.
- *RKTM*³⁵ is a recurrent knowledge tracing machine model by temporally enriching the encoding of KTM and DAS3H with the knowledge state of students.
- *AKT*¹⁶ is an attentive KT model, it builds context-aware representations of questions and utilizes a novel monotonic attention mechanism to relate a learner's current answers to questions to their past responses.
- *CoKT*⁴² enriches the question information into the KT model using a question-question collaborative embedding method.
- *IEKT*³⁹ is a single-state RNN-based model. It incorporates the learner characteristics into the model by considering learners' cognition levels and knowledge acquisition sensitivity on questions.

Among the baseline models, the former three are skill-based models, in which the KT is based on the skills contained in the questions. The latter 10 and the proposed model are question-based models that account for the distinctive question information.

5.4 | Setup and implementation

Before conducting the experiments, we extracted 20% of the sequences in the data set as the test set and retained the remaining 80% as the training set.

To embed the nodes in the graph using Metapath2Vec, we set the length of all meta-paths as $\varphi = 7$ and the number of paths as $\aleph = 100$ for each question node in the graph. The embedding dimension d of the skill and question representations was set to 128. We pretrain these embeddings in the graph and obtain the stable results based on Equation (7), and fix them in the training and testing process. The final dimension of the convolutional question representation was $d' = 256$. The size of the hidden layers of the LSTM was set to 256. The other hyperparameters were set through grid searching. The embedding matrix of the correctness and

all parameter matrices in the networks were randomly initialized and updated through the training process.

The model was optimized using Adam optimization of the learning rate on a case-by-case basis in the three data sets. The norm clipping threshold and batch size were maintained at 10 and 64, respectively. Similarly to the existing models, the sequence length of the model input was fixed at 200. Accordingly, the long sequences were divided into several short sequences and the short sequences were padded with null symbols to extend their length to 200.

The proposed model was implemented using TensorFlow. Our model was tested over 50 epochs because it converged over that period. The other baselines were implemented with their best parameter settings, as specified in the original works.

As the evaluation metric, we selected the area under the receiver operating characteristic (ROC) curve (AUC), which is widely used in existing studies. An AUC of 0.5 indicates that the result is no better than random guessing; the larger the value, the better is the model performance.

6 | RESULTS AND ANALYSIS

This section presents the experimental results and model analysis. To answer our first research question in Section 5.1, Section 6.1 compares the learner-score prediction performances of our proposed KSGKT model and all baseline models. The other five questions are answered in relevant model analyses. In Section 6.2.1, the contribution of the embedding learning on the KS-enhanced graph is evaluated in an ablation study and the second question is answered. Section 6.2.2 presents the visualization of the question embedding learned on the graph and answers the third question. Sections 6.2.3 and 6.2.4 compare the eight methods for KS discovery and verify the interpretability of the inferred KS, and the fourth question is answered. To answer the fifth and sixth research questions, the contributions of two separate components (convolutional question representation and the attention mechanism) are evaluated through the ablation studies in Sections 6.2.5 and 6.2.6.

6.1 | Performance prediction

The different models were evaluated by their performances in predicting the future learner scores from the estimated knowledge state. Table 6 presents the AUC results of all models on the three data sets.

Our model outperformed the other models on all three data sets. Specifically, the AUC scores of the KSGKT model were 0.8242, 0.7851, and 0.7754 on the Assist0910, Assist1213, and EdNet data sets, respectively, 3.97%, 1.39%, and 2.25%, respectively, above those of the state-of-the-art AKT model. Similarly to the original DKT model, our model processes the time-series data using a recurrent neural network framework, but achieved 8.3%, 5.95%, and 8.65% higher AUCs than the DKT model on the Assist0910, Assist1213, and EdNet data sets, respectively.

The skill-based BKT model was the worst performer among the models because it tracks the mastery of each skill separately, without considering a contextual trial sequence of all skills. DKT and DKVMN achieved similar performances and considerably outperformed the

TABLE 6 Comparisons of the AUC results of different models on the three data sets

	Model	ASSIST0910	ASSIST1213	EdNet
Skill-based model	BKT	0.6571	0.6204	0.6027
	DKT	0.7412	0.7256	0.6889
	DKVMN	0.7559	0.7247	0.6921
Question-based model	KTM	0.7582	0.7212	0.6899
	DKT-Q	0.7306	0.7158	0.6812
	DKT-Q&S	0.7616	0.7389	0.7235
	DKT-CQE	0.7998	0.7686	0.7523
	GIKT	0.7845	0.7712	0.7529
	DAS3H	0.789	0.741	0.731
	RKTM	0.7650	0.7690	–
	AKT	0.8152	0.7720	0.7658
	CoKT	0.7928	0.7802	0.7464
	IEKT	0.7720	0.7341	0.7305
	KSGKT	0.8242*	0.7851*	0.7754*

BKT model, confirming the effectiveness of applying deep neural networks to this task; however, they were slightly outperformed by the other question-based models. KTM framework, which incorporates several traditional models, typically obtained similar AUC scores to those of DKT and DKVMN. DKT extended with various input-question embeddings (DKT-Q, DKT-Q&S, and DKT-CQE) demonstrated noticeable performance differences. DKT-Q using the one-hot encoding of question representations performed much worse than the original DKT model, owing to the sparsity of question interactions in these data sets. DKT-Q&S and DKT-CQE decidedly outperformed the original DKT and DKT-Q models, consistent with our intuition that each question contains distinctive information even when it requires the same skills as one or more other questions in the data set. Therefore, incorporating the distinctive question and skill information into the question representations can improve the model performance. The comparison between DKT and DKT-CQE also shows the effectiveness of the proposed convolutional question representation. Moreover, the improvements in the AUC scores of these DKT-extended models were smaller on Assist1213 than on the other two data sets. As Assist1213 is a single-skill data set, incorporating the skill information into the question representation provides less additional information on this data set than on the other data sets. DAS3H and RKTM are built on the KTM model by considering additional rich information, and show better results than the original KTM model. Here we do not present the results of RKTM on the EdNet as the code for implementing RKTM is not published by the author. IEKT also shows better performance than the DKT model by considering the individual learner's characteristics in the KT procedure. Our KSGKT model outperforms GIKT, AKT, and CoKT, which also take the well-designed question embedding as input. This result validates the effectiveness of the proposed KS-enhanced question-skill graph learning method.

6.2 | Model analysis

In this section, our proposed model is further investigated in four additional experiments. Specifically, we investigate the effects of the embedding learning on graphs and the convolutional question representation, compare different methods of inferring KS from data, and compare the performances of the three attention-calculation methods.

6.2.1 | Effect of embedding learning on graph

Besides the aforementioned three extensions of DKT models, we extended our KSGKT model with the following question embeddings:

- *KSGKT-Q*: In this embedding, we removed the module of embedding learning on the graph and inputted the one-hot question embedding to our KSGKT model.
- *KSGKT-S*: Similar to the KSGKT-Q embedding, but here we inputted the one-hot skill embedding to our KSGKT model.
- *KSGKT-Q&S*: Similar to the KSGKT-Q embedding, but here we concatenated the question and skill embedding and inputted the result to our KSGKT model.
- *KSGKT-CQE*: This model was the proposed model, renamed to emphasize that we learned the convolutional question embedding from the graph and inputted the result to our KSGKT model.

The AUC performances of the different question-embedding methods in the DKT, GIKT, and the proposed KSGKT models are displayed in Table 7. Herein, we mainly compared the one-hot embedding, the GNN-based graph embedding in GIKT, and the KS-enhanced graph embedding in the proposed KSGKT model. The one-hot embeddings in the DKT-extended models (DKT, DKT-Q, and DKT-Q&S) and the KSGKT-extended models (KSGKT-Q, KSGKT-S, and KSGKT-Q&S) were outperformed by the other two embedding methods, validating that the embeddings learned from the graph incorporate more meaningful information about the graph (i.e., the relations among the questions and skills). Moreover, the KSGKT-extended models usually outperformed the

TABLE 7 Comparisons of different question-embedding methods in different models on the three data sets

Methods	ASSIST0910	ASSIST1213	EdNet
DKT	0.7412	0.7256	0.6889
DKT-Q	0.7306	0.7158	0.6812
DKT-Q&S	0.7616	0.7389	0.7235
DKT-CQE	0.7998	0.7686	0.7523
GIKT	0.7845	0.7712	0.7529
KSGKT-Q	0.7409	0.7277	0.6987
KSGKT-S	0.7523	0.7399	0.7043
KSGKT-Q&S	0.7682	0.7464	0.7293
KSGKT-CQE	0.8242*	0.7851*	0.7754*

*The best results.

corresponding DKT-extended models because they employ an attention mechanism. Comparing the performances of the DKT-CQE and DKT-extended models (also the KSGKT-CQE and the KSGKT-extended models), we validated the effectiveness of the KS-enhanced graph embedding method and its ability to improve the existing models. Comparing the performances of the GIKT and KSGKT-CQE models, we further confirmed that embedding learning on our KS-enhanced graph outperformed embedding learning on the original question–skill graph.

6.2.2 | Visualization of question embedding learned on a graph

Figure 5 illustrates the question embeddings learned from the original and KS-enhanced graphs of Assist0910 and EdNet. For visual clarity, we randomly selected 50 skills and their corresponding questions in the two data sets and scattered the embedding vectors of these questions in the embedding matrix learned using Metapath2Vec. Specifically, we visualized the high-dimensional data by projecting the high-dimensional embedding vectors of the questions onto two-dimensional points. The projection was implemented using t-SNE⁵⁶ in Python.

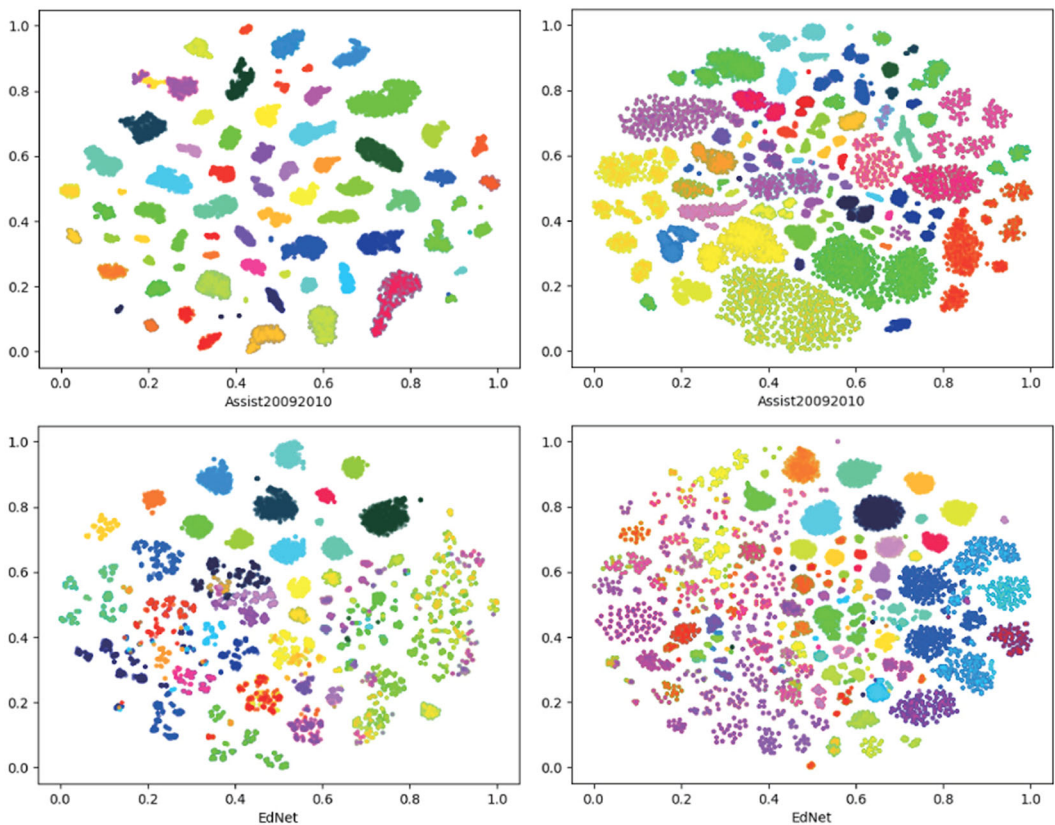


FIGURE 5 Embeddings of question nodes learned from the graphs of the Assist0910 and EdNet data sets. Each panel presents the questions associated with 50 kinds of skills in different categories (labeled with different colors). The upper left and right panels show the question embeddings learned from the original question–skill graph and the KS-enhanced graph using the adjusted Kappa in Assist0910, respectively. The lower figures show the question embeddings learned from the corresponding graphs in EdNet [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.22763)]

As an example, we enhanced the original question–skill graph using the adjusted Kappa in Assist0910. As shown in Figure 5, the question embeddings of both data sets were highly structured: questions requiring the same skill tended to be clustered while questions belonging to different categories were well separated. Moreover, the question embeddings of the original graphs of both data sets were well separated into different categories, unlike the distributions in the KS-enhanced graph. This result is consistent with our motivation, as the KS is intended to bring the skill domain into the enhanced graph. Therefore, as each question is potentially associated with more skills (the associated skills plus the prerequisite skills), the distances between questions in related categories became much closer. Also notable are the different distributions of the question embeddings learned from the original graphs of the Assist0910 and EdNet data sets. The boundaries of the questions requiring different skills were much clearer in the Assist0910 data set than in EdNet. This result can be traced to the lower average number of skills related to each question in Assist0910 (1.207) than in EdNet (2.276). Therefore, a question in EdNet more likely requires multiple skills so the intergroup distances were much closer in the question embeddings of this data set.

6.2.3 | Comparison of KS inferences by different methods

In Section 4.1, we presented eight methods for inferring KS from the response data. This section compares the performances of these methods.

The number of relations in the inferred KS can be controlled by a threshold. Here, the threshold for each method on the three data sets was decided on a case-by-case basis because the different coefficients have different ranges. Figure 6 shows the numbers of relations in KS with different thresholds after applying the Cohen Kappa, adjusted Kappa, and Phi coefficient on the Assist0910 and EdNet data sets. In our experiments, we set the threshold of both data

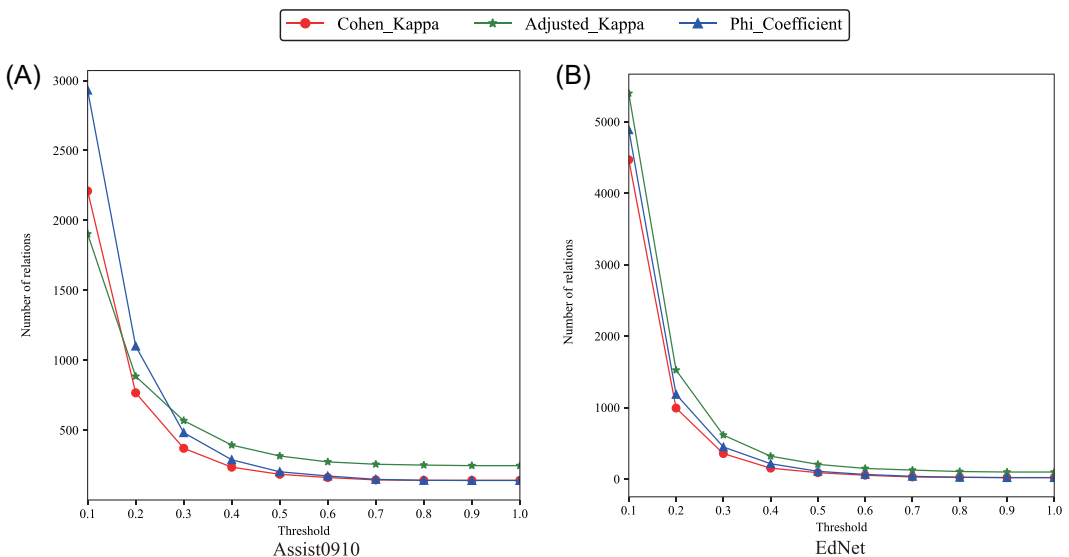


FIGURE 6 Number of inferred relations in KS using the three methods with different thresholds on two data sets. (A) Assist0910 and (B) EdNet [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.22763)]

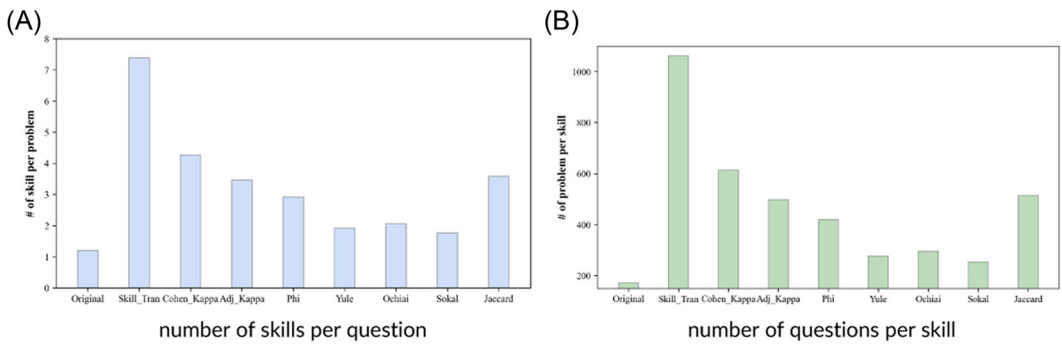


FIGURE 7 Average number of skills per question and average number of questions per skill before and after considering the KS on the Assist0910 data set; (A) number of skills per question; (B) number of questions per skill [Color figure can be viewed at wileyonlinelibrary.com]

sets to 0.2 to obtain a proper number of relations for the KS. Figure 7 compares the average number of skills per question and the average number of questions per skill before and after considering the KS on Assist0910. The original numbers of skills per question and questions per skill were both very low, but after incorporating KS with the questions using various methods, both values increased by different degrees, indicating that more meaningful information was included in the question representation.

Figure 8 compares the AUC results of the predictions of nine methods on the three data sets. Here the “original” method represents the embedding learning on the original question–skill relation graph and the other eight methods are based on the KS-enhanced graph. As evidenced in the figure, the eight KS enhanced methods generally outperformed the “original” method on all three data sets, validating the effectiveness of the KS enhanced graph in KT tasks. The adjusted Kappa yielded the best performance on both Assist0910 and EdNet, whereas the skill transaction method performed best on Assist1213.

6.2.4 | Visualization of inferred KS

Figure 9 illustrates the KS graphs of four methods, inferred from the learner response data in Assist0910. The right-hand side of this figure enlarges a part of the graphs to show their local connections. The nodes and edges in the KS form dense graphs with similar structures, showing several interconnected nodes. These graphs also show some interesting properties. In the adjust-Kappa graph, four nodes (25, 111, 90, and 95) were locally interconnected and revealed a perfect ordering of the skills (prerequisite and postrequisite relations) in the geometry. The local connections in the Phi coefficient graph also presented reasonable relations among the three skills. These results confirm that our KS discovery methods can infer prerequisite skill pairs from the ordering of learners' mastery of skills.

6.2.5 | Effect of convolutional question representation

To evaluate the effectiveness of the convolutional question-representation module, we fed the following question representations into our KSGKT model:

- *KSGKT with only QE*: The question embedding learned from the graph was input to our KSGKT model.
- *KSGKT with only SE*: The skill embedding learned from the graph was input to our KSGKT model.
- *KSGKT with QE&SE*: The question and skill embeddings learned from the graph were concatenated and input to our KSGKT model.

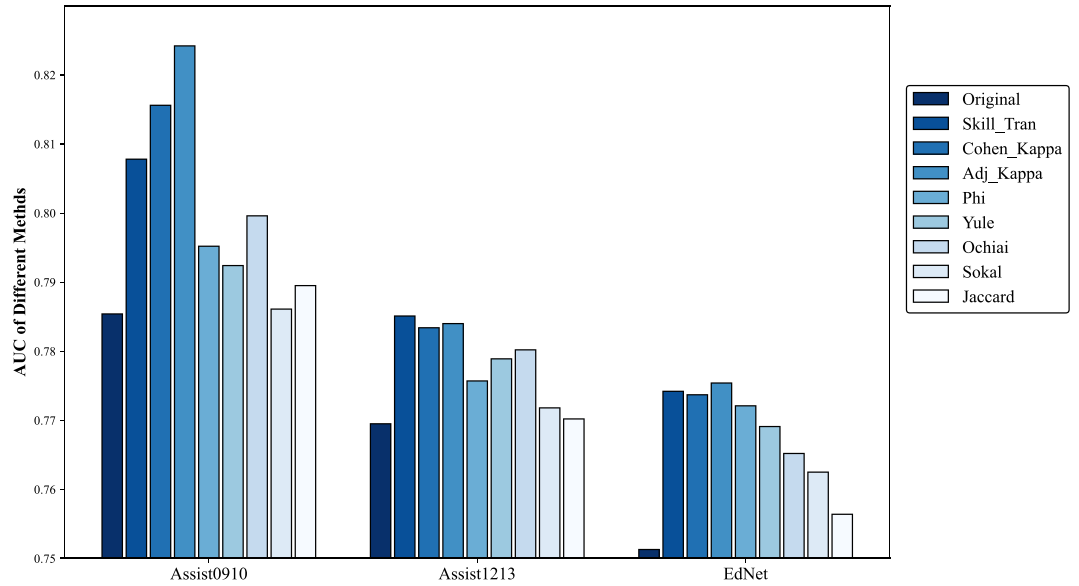


FIGURE 8 KS inferred from the learner response data [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.2763)]

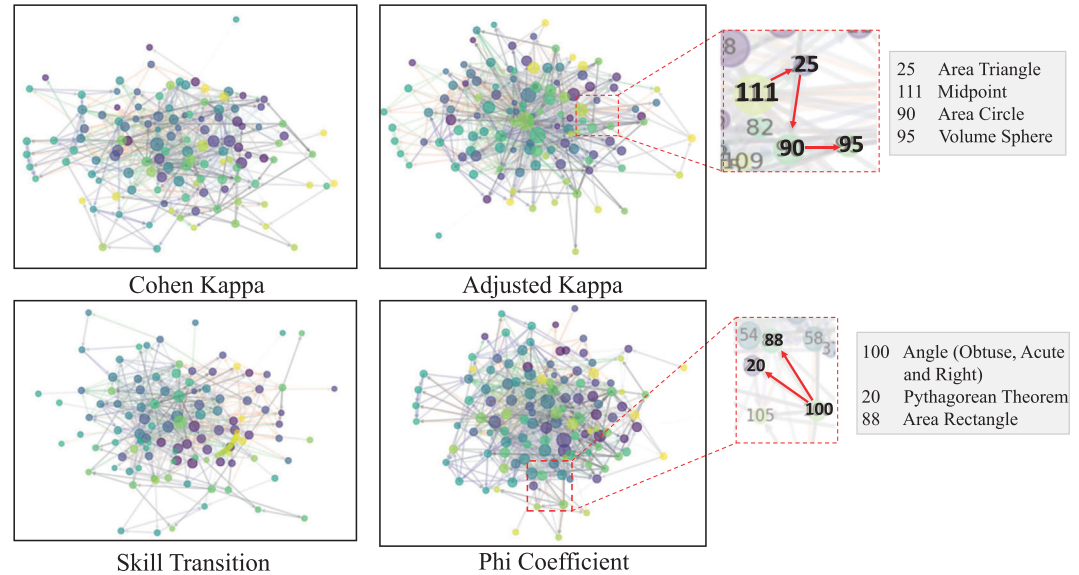


FIGURE 9 Visualization of the directed KS graphs generated by four methods on Assist0910 [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.2763)]

TABLE 8 Comparisons of AUC results of various KSGKT extended models with different question representations on the three data sets

Methods	ASSIST09	ASSIST12	EdNet
KSGKT with only QE	0.8005	0.7636	0.7489
KSGKT with only SE	0.8047	0.7569	0.7543
KSGKT with QE&SE	0.8132	0.7764	0.7603
KSGKT with CQE	0.8242*	0.7851*	0.7754*

*The best results.

- *KSGKT with CQE*: This was the proposed model, renamed to emphasize that the convolutional question embedding learned from the graph was input to our KSGKT model.

Table 8 compares the performances of the KSGKT extended models. The KSGKT with the convolutional question embedding achieved the best results on all three data sets, indicating the effectiveness of the convolutional question-representation module. The KSGKT with QE&SE obtained the second best results, and the KSGKT with SE alone generally outperformed the KSGKT with QE alone (the exception was the Assist1213 data set). The convolutional question representation acquires the distinctive features of the questions and also their interactions, thus improving the performance of the proposed model.

6.2.6 | Effects of the three attention-calculating methods

Table 9 compares the performances of the three attention methods on the data sets. The proposed method with the combined attention method far outperformed the proposed method with the shared skill-based and question similarity-based methods, indicating that incorporating the knowledge states of many related questions improved the model performance. However, the models with all three attention-calculating models significantly outperformed the model without attention, confirming that the attention mechanism benefits the KT task. This result is consistent with existing work,^{15,16,41} which reported that learners' past experience on related questions affects the performance on the current question.

7 | APPLICATIONS

The results of knowledge tracing and knowledge structure discovery are promising for various applications that can improve the learning and teaching efficiency of online systems. As examples, we introduce the following applications.

7.1 | Fine-grained diagnostic report

Most of the existing tutoring systems provide learners only with coarse-grained information such as correct/incorrect feedback or the score/rank of their exercise process.^{13,27} From a tutoring viewpoint, learners who understand the strengths and weaknesses of their knowledge

TABLE 9 Comparison of AUC results of the proposed model with three attention-calculating methods, evaluated on the three data sets

Methods	ASSIST0910	ASSIST1213	EdNet
Without attention	0.7998	0.7686	0.7523
shared skill-based attention	0.8229	0.7784	0.7729
question similarity-based attention	0.8233	0.7839	0.7727
combined attention	0.8242*	0.7851*	0.7754*

*The best results.

points can remedy these weaknesses and improve themselves through self-regulated learning. From a teaching viewpoint, a comprehensive diagnostic report would help teachers identify the knowledge levels of both the whole class and individual students. Based on this information, they can design and provide timely interventions of the learning procedures.

Our model can easily generate a fine-grained diagnostic report and provide the dynamic evolution of learners' mastery of each skill over time. Figure 10 shows the procedure of obtaining the dynamic skill mastery of learners. When a learner attempts a series of questions, the hidden layer of the network retains the relevant information in the learners' exercising history. Intuitively, this knowledge retention can be considered as an embedding of the knowledge states of the learner⁷ (shown as h_i). As the hidden vector and the skill embedding representation have the same dimension, we can map the hidden knowledge states vector and the skill embedding vector to the same space. The mastery level of each skill is then computed as

$$m(s_k) = \sigma(h_i \odot s_k) \in [0, 1],$$

where $\sigma(u) = 1/(1 + \exp(-u))$ is the sigmoid function and $\odot$ refers to the inner product. The right-hand side of Figure 10 illustrates the mastery levels of a learner on five skills in the Assist0910 data set. Note that the skill mastery steadily evolved as the learner gradually mastered all five skills after attempting 21 questions. Assisted by this fine-grained diagnostic report, learners can focus on their weak knowledge without repeated training on their already mastered skills. This enlightenment will greatly improve students' learning efficiency.

7.2 | Individualized learning content

KT is essential for adaptive learning and might also increase the intelligence of ITS.^{4,11} Based on the obtained knowledge mastery, adaptive services such as remedial learning materials that meet learners' individual needs can be automatically provided.^{11,27} Besides obtaining the required and prerequisite skills for solving a problem, our question-representation method infers the cognitive difficulty of the question from a learner's exercising history. Content predicted to be incompatible with the learners' knowledge level (too easy or too difficult) can be skipped or delayed, thereby effectively improving the students' learning paces while maintaining their engagement.¹¹ Many tutoring systems use similar exercises to identify whether a learner has mastered a certain type of exercise.²⁷ Our proposed method can be easily implemented for this purpose. In the question embeddings of the question-skill graph, questions linked by related skills are much closer in the embedding space than questions requiring different skills; hence,

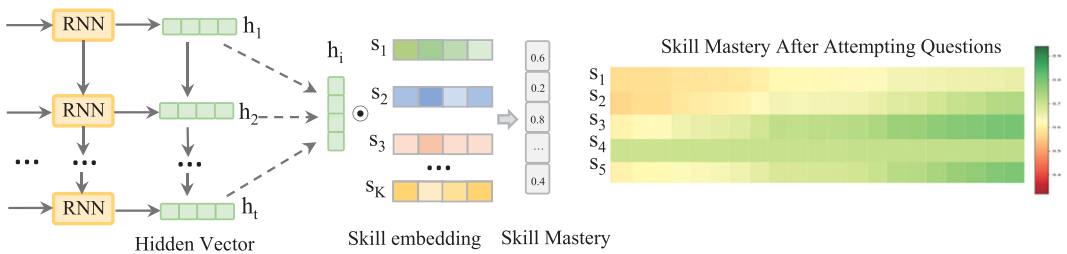


FIGURE 10 Obtaining a learner's knowledge state of each skill through interaction between the hidden knowledge states in the network and skill representation. The heat map on the right shows the evolution of the learner's skill mastery before and after attempting a series of questions. Each cell in the map indicates the learner's mastery of a specific skill at some time point. Over time, the mastery level increases from low (orange) to high (green) as the learner gradually masters the skills. RNN, recurrent neural network [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.22763)]

similar questions can be filtered from the question bank by performing simple similarity calculations on their embedding representations.

7.3 | Adaptive learning-path recommendation

Adaptive learning-path recommendation reasonably arranges the order of the learning contents to generate a well-defined learning path. Along this path, a learner can efficiently complete the learning target and alleviate the information overload issues in e-learning.⁵⁷ As pointed out in,²² the existing methods consider either the knowledge level or knowledge structure, but not their combination. In contrast, our model obtains the learners' knowledge level and automatically discovers the knowledge structure of the domain. Incorporating both parts, we can provide an adaptive learning-path recommendation service. In the example of Figure 11, the learner is requested to master skill "E" along the recommended learning path "B → A → C → E" based on his or her current knowledge level. This path follows the logicity determined by the knowledge structure. As the learner has already mastered skill "B," he or she approaches skill "C" from skill "B" rather than from skill "D." Along the learning path, the learner is provided with the corresponding learning contents. After completing this self-learning process, the learner has gradually mastered all skills along the learning path. Such self-awareness can greatly improve the adaptive navigation ability of existing ITS.

8 | CONCLUSIONS AND FUTURE WORK

This paper presented a knowledge structure-enhanced graph representation learning model for knowledge tracing (KSGKT) with an attention mechanism. By incorporating the knowledge structure into the knowledge tracing model, the model dynamically traces the learner's knowledge proficiency, thus alleviating the sparseness of the interaction data and the neglect of distinctive information related to the questions themselves and their relations. These problems severely limit the efficacies of previous skill-based models.

To automatically obtain the KS in the domain, we first explored the abilities of eight methods to infer the domain KS from learner response data based on the mastery orders of

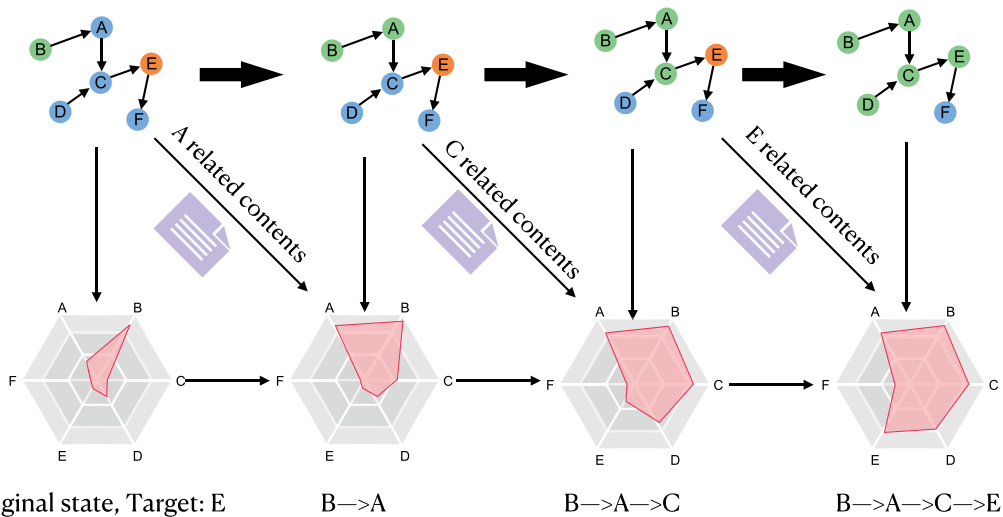


FIGURE 11 Adaptive learning-path recommendation based on the learners' knowledge level and the knowledge structure. The learning path “ $B \rightarrow A \rightarrow C \rightarrow E$ ” is recommended to a learner for mastering the target skill E. The green and orange nodes in the knowledge graph represent the already mastered skills and the target skills, respectively. The below-radar graphs show the changes in the learner's knowledge states on each skill. Gradually, the learner mastered all skills in the learning path [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/int.22763)]

pairs of skills. After integrating the KS with the KT procedure, we leveraged a graph-representation learning model and obtained the question and skill embeddings from the KS-enhanced graph. To incorporate more distinctive information regarding the questions, we extracted the adaptive cognitive question difficulty from the learning histories of individual learners, and proposed a convolutional representation method that fuses the question difficulty with the question itself and its associated skill embedding. We thus obtained a comprehensive representation of each question. Feeding these representations into the proposed KT model with an attention mechanism, we can predict the learning performance on new problems. Extensive experiments conducted from six perspectives on three real-world data sets demonstrated the superiority and interpretability of our model for learner performance modeling and KS discovery, validating its potential applicability to real educational environments.

In future work, we intend to embed KS learning into our model in an end-to-end manner. The KS will then be automatically learned in the training process rather than computed from the learner performance data. Such automation can potentially expand the KS to new skills in the domain, for example, enlarging the KS of mathematics from the primary school to high school level. Moreover, we plan to integrate KT in real educational scenarios, providing students with proper adaptive remedial materials based on their individual needs. In case studies, we will verify the effectiveness of our method in online learning.

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ENDNOTES

* Assist0910: <https://sites.google.com/site/assistentmentsdata/home/assistentment-2009-2010-data/skill-builder-data-2009-2010>

† Assist1213: <https://sites.google.com/site/assistentmentsdata/home/2012-13-school-data-with-affect>

‡ EdNet: http://bit.ly/ednet_kt1

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